

Output

enter first number

5

enter second number

3

choose operation

1. addition

2. Subtraction

3. Multiplication

4. division

1. n

addition is = 8

Aim:- Program to implement case structure, use of break & continue

(a) Write a shell script to make a basic calculator which performs addition, subtraction, multiplication, division.

echo enter first edition number

read a

echo enter second number

read b

echo choose operation

echo 1. addition

echo 2. Subtraction

echo 3. multiplication

echo 4. Division

read chose

case \$choice in

1) result = \$((\$a+\$b))

echo addition is = \$result;;

2) result = \$((\$a-\$b))

echo subtraction is = \$result;;

3) result = \$((\$a*\$b))

echo multiplication is = \$result;;

4) result = \$((\$a/\$b))

echo division is = \$result;;

*) echo Invalid choice!;;

esac

Output

Select month

1. January and February
2. March & April
3. May and June
4. July
5. August
6. September & October
7. November & December

1.

Next 4 months that have 31 days are

March 2025

Su	Mo	Tu	We	Th	Fr	Sa
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

July 2025

Su	Mo	Tu	We	Th	Fr	Sa
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

May 2025

Su	Mo	Tu	We	Th	Fr	Sa
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

August 2025

Su	Mo	Tu	We	Th	Fr	Sa
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

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(b) Write a shell script to print starting 4 months having 31 days.

echo select month

echo 1: January and February

echo 2: March & April

echo 3: May and June

echo 4: July

echo 5: August

echo 6: September & October

echo 7: November & December.

read choice

case \$choice in

1) echo Next 4 months that have 31 days

cal 03 2026

cal 05 2026

cal 07 2026

cal 08 2026 ;;

2) echo Next 4 months that have 31 days

cal 05 2026

cal 07 2026

cal 08 2026

cal 09 2026 ;;

3) echo Next 4 months that have 31 day are

cal 07 2026

cal 08 2026

cal 10 2026

cal 12 2026 ;;

Teacher's Signature _____

Output

Value = 0

Value = 1

Value = 2

Value = 3

Value = 4

output

value = 0

value = 1

value = 2

value = 3

value = 4

value = 5

value = 6

value = 7

value = 8

value = 9

value = 10

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4) 4) echo Next 4 months that have 31 days

cal 08 2026

cal 10 2026

cal 12 2026

cal 01 2026 ;;

5) echo Next 4 months that have 31 days

cal 10 2026

cal 12 2026

cal 01 2026

cal 03 2026 ;;

6) echo Next 4 months that have 31 days

cal 12 2026

cal 01 2026

cal 03 2026

cal 05 2026 ;;

7) echo Next 4 months that have 31 days

cal 01 2026

cal 03 2026

cal 05 2026

cal 07 2026 ;;

esac

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(c) Use of Break Statement

```
i = 0
while [ $i -lt 10 ]
do
  echo value = $i
  i = $(($i+1))
  if [ $i -eq 5 ]
  then
    break
  Fi
done.
```

(d) Use of Continue

```
i = 0
while [ $i -lt 10 ]
do
  i = $(($i+1))
  if [ $i -eq 5 ]
  then
    continue
  Fi
  echo value = $i
done.
```


Viva Question

Q1 What is the use of case statement?

Ans A case statement is used to handle multiple condition efficiently.

Q2 How to terminate case statement in shell scripting?

Ans A case statement is terminated using esac, which is case spelled backward.

Q3 What is the generic case?

Ans The generic case refers to the default case that executes when no specified patterns match the input. It is represented by *.

Q4 What do you mean by break statement?

Ans The break statement is used to exit a loop prematurely when a specific condition is met.

Q5 What do you mean by continue statement?

The continue statement is used to skip the current iteration of a loop and move to next iteration without exiting the loop entirely.

Output A

Enter A Number

153

Number is armstrong number

Date _____

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Aim:- Write a program to implement function.

(A) Write a shell script to find a number is Armstrong or not.

```
is_armstrong() {
```

```
  n=$1
```

```
  sum=0
```

```
  while [ $n -gt 0 ]
```

```
  do
```

```
    rem=$((n%10))
```

```
    sum=$((sum + (rem * rem * rem)))
```

```
    n=$((n/10))
```

```
  done
```

```
  if [ $n -eq $sum ]
```

```
  then
```

```
    echo Number is armstrong number
```

```
  else
```

```
    echo Number is not armstrong number
```

```
  fi
```

```
}
```

```
echo Enter A Number
```

```
read n
```

```
is_armstrong $n
```

Teacher's Signature _____

Output B

Enter a Number

123

No it's not a palindrom number.

Date _____

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(B) Write a shell script to find a number is
palindrome or not.

```
is_palindrome() {  
    sum = 0  
    m = $n  
    while [ $n -gt 0 ]  
    do  
        rem = ${n%10}  
        sum = $((($sum * 10) + $rem))  
        n = ${n/10}  
    done  
    if [ $m -eq $sum ]  
    then  
        echo it's a palindrome number  
    else  
        echo no it's not a palindrome number  
    fi  
}  
echo Enter a Number  
read n  
is_palindrome $n
```

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Output 5-

Enter no. of terms

8

Fibonacci Series

0

1

1

2

3

5

8

13

Date _____

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(C) Write a shell script to print Fibonacci series

```
is-fibonacci() {
```

```
    a=0
```

```
    b=1
```

```
    count=0
```

```
    echo Fibonacci series
```

```
    while [ $count -lt $n ]
```

```
    do
```

```
        echo $a
```

```
        Fn=$((a+b))
```

```
        a=$b
```

```
        b=$Fn
```

```
        count=$((count+1))
```

```
    done
```

```
}
```

```
echo Enter no. of terms
```

```
read n
```

```
is-fibonacci $num
```

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output D

Enter A Number

97

No is prime.

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(D) Write a shell script to find prime numbers.

```
is_prime () {  
    n = $1  
    if [ $n -lt 2 ]  
    then  
        echo not a prime number  
    fi  
    i = 2  
    while [ $(($i * i)) -le $n ]  
    do  
        if [ $(($n % i)) -eq 0 ]  
        then  
            echo no is not prime  
            return  
        fi  
        i = $(($i + 1))  
    done  
    echo no is prime  
}  
echo Enter A Number  
read n  
is_prime $n
```

Teacher's Signature _____

Viva Question

Q1 What is a function?

Ans A function is a reusable block of code that performs a specific task.

Q2 How to call a function in a shell script?

Ans

```
my_function () {  
    echo Hello  
}  
  
my_function
```

Q3 What are the advantages of using functions?

Ans Code reusability, Improves readability, Easier debugging, Modular approach.

Q4 What is the difference between local & global variable?

Ans Global variables are accessible anywhere.
Local variables are limited to the function where they are defined.

Q5 What do you mean by Palindrome number?

Ans A palindrome number is a number that can read the same forward & backward.